

ADAPTIVE DIGITAL SAT PRACTICE TEST

Horizon Series — Form A

This practice test mirrors the adaptive structure of the Digital SAT. Section 1 (Reading & Writing) contains Module 1 (27 questions for all students), then branches into an Easier Module 2 or Harder Module 2 based on Module 1 performance. Section 2 (Math) follows the same adaptive structure with 22 questions per module.

Test Structure:

- **Section 1 Module 1:** 27 Reading & Writing questions (all students)
- **Section 1 Module 2 Easier:** 27 questions (~70% Easy/Medium, ~30% Hard)
- **Section 1 Module 2 Harder:** 27 questions (~30% Easy/Medium, ~70% Hard)
- **Section 2 Module 1:** 22 Math questions (all students)
- **Section 2 Module 2 Easier:** 22 questions (~70% Easy/Medium, ~30% Hard)
- **Section 2 Module 2 Harder:** 22 questions (~30% Easy/Medium, ~70% Hard)
- **Total: 147 original questions**

SECTION 1: READING AND WRITING

MODULE 1 (Same for all students — 27 Questions)

Question 1 — Skill: Words in Context | Difficulty: Easy

The deep-sea drilling campaigns of the 1960s and 1970s recovered sediment cores from ocean floors worldwide, generating vast geological datasets. Researcher Tanya Mikhailova, who studies ancient volcanic eruptions recorded in these sediments, continues to analyze the drill samples, confirming that the expeditions' contributions to earth science are _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) isolated B) provisional C) unprecedented D) enduring
-

Question 2 — Skill: Words in Context | Difficulty: Easy

Because diatoms and other phytoplankton—photosynthetic microorganisms drifting in sunlit waters—are consumed by zooplankton as well as by juvenile fish and shellfish, phytoplankton are _____ pelagic food chains. Reduced phytoplankton density, which may be triggered by nutrient depletion, thus has a cascading negative effect on open-ocean ecosystems.

- A) tangential to B) indistinguishable from C) fundamental to D) competitive with
-

Question 3 — Skill: Words in Context | Difficulty: Easy

Certain novels invite dramatically varied readings—the works of Haruki Murakami can _____ as many distinct analyses as there are critics willing to engage with them—and provided that a reader's interpretation is not rooted in carelessness or willful distortion, it is difficult to claim that the reader has fundamentally misread the text.

- A) dismiss B) obscure C) constrain D) generate
-

Question 4 — Skill: Words in Context | Difficulty: Easy

The color vocabularies of many indigenous languages in the Amazon Basin tend toward the _____: the Piraha language of Brazil has been analyzed as containing only rough equivalents for light and dark, and the Dani language of Papua uses just two basic color terms, while the global average per language exceeds eleven.

- A) sparse B) rigid C) uniform D) elaborate

Question 5 — Skill: Main Purpose | Difficulty: Medium

Established in Edinburgh, Scotland, in 1826, the Royal Scottish Academy is devoted to promoting contemporary visual art produced by Scottish artists and artists working in Scotland. Since its founding, it has mounted over two thousand exhibitions of paintings, sculpture, and architecture. More recently established Scottish institutions dedicated to visual art include the Dundee Contemporary Arts center. Opened in Dundee in 1999, it focuses on international contemporary art with a particular emphasis on film and printmaking.

Which choice best states the main purpose of the text?

- A) To trace a historical development that led to the founding of two cultural organizations
- B) To compare the funding models of two Scottish art institutions
- C) To present information about two institutions, including each institution's area of focus
- D) To argue that one institution has a larger collection than the other

Question 6 — Skill: Main Purpose | Difficulty: Medium

The following text is adapted from William Cullen Bryant's 1821 poem "Green River." The poem is a tribute to a river near the poet's hometown in western Massachusetts.

Though forced to drudge for the dregs of men,
And scrawl strange words with the barbarous pen,
And mingle among the jostling crowd,
Where the sons of strife are subtle and loud—
I often come to this quiet place,
To breathe the airs that ruffle thy face,
And gaze upon thee in silent dream.

Which choice best states the main purpose of the text?

- A) It describes the speaker's daily routine of visiting the river for exercise.
- B) It contrasts the peacefulness of the river with the harshness of the speaker's working life.
- C) It emphasizes the danger of the river during storm season.
- D) It compares the appearances of two different rivers.

Question 7 - Skill: Detail (According to the Text) | Difficulty: Medium

In northern Australia, the Yolngu people operate the Buku-Larrngay Mulka Centre, a community-controlled art center and museum. Drawing on traditional law and ceremony to guide its exhibitions, this institution presents Yolngu cultural heritage to visitors on the community's own terms. The Tiwi Design collective on Bathurst Island employs a similar community-led approach in its own cultural space. Both centers contrast with museums that are not community-controlled; when displaying Indigenous Australian artifacts, such museums tend to anticipate primarily non-Indigenous visitors and rely on Western curatorial frameworks for organizing displays.

According to the text, what is one way that non-community-controlled museums typically differ from the cultural centers operated by the Yolngu and Tiwi communities?

- A) The museums typically display fewer artifacts in their exhibitions.
- B) The museums are generally smaller in physical size.
- C) The museums are primarily designed for non-Indigenous audiences.
- D) The museums focus exclusively on historical rather than contemporary art.

Question 8 — Skill: Inference / Complete the Text | Difficulty: Medium

The Galápagos archipelago is home to the flightless cormorant (*Nannopterum harisi*), one of the world's rarest seabirds. Due to the islands' distinctive marine and terrestrial conditions, the flightless cormorant became highly specialized as it adapted to forage in the nutrient-rich coastal waters surrounding the volcanic islands. However, like many highly specialized species, the flightless cormorant is particularly vulnerable to environmental disruptions, such as El Niño events, oil spills, and introduced predators, that upset the delicately balanced ecosystems on which the birds depend.

Based on the text, why is the flightless cormorant a highly specialized species?

- A) It lives exclusively in the deepest coastal waters surrounding the Galápagos
- B) It evolved to thrive in a particular habitat found in the Galápagos archipelago.
- C) It adapted over time to share its habitat with other specialized seabird species.
- D) It inhabits an ecosystem that faces numerous environmental threats.

Question 9 — Skill: Main Idea | Difficulty: Medium

The following text is adapted from Willa Cather's 1913 novel *O Pioneers!* The narrator describes a winter landscape on the Nebraska prairie.

The main street of the little town was deserted. The wind had already begun to pile the snow into drifts along the board sidewalks. The few pedestrians astir wrapped themselves to the eyes. I drove along the ridge of the Divide, looking down at the brown, frost-bitten fields below. The land was so flat that one could see the distant church steeples. The frozen country seemed to press

in upon me, yet as the road turned south the wind fell and the prairie opened out before me, wide and still, like a great sheet of iron under the grey sky.

Which choice best states the main idea of the text?

- A) The narrator contrasts feelings of unease with a growing sense of spaciousness as the landscape shifts along the drive.
- B) The narrator is eager to escape the cold weather and is frustrated by the length of the journey.
- C) Although the narrator did not plan to travel through the countryside, the experience proves exhilarating.
- D) The narrator observes that the town and the surrounding prairie are equally desolate in winter.

Question 10 — Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Graph: A line graph titled “Mean Wetland Area for Three Habitat Classifications in the Murray-Darling Basin, Australia” shows years 1975–2015 on the x-axis, mean wetland area (hectares) 0–200 on the y-axis. Class A (solid/triangles): ~180 in 1975, drops to ~40 by 1995, stays ~50. Class B (dashed/squares): ~100 in 1975, drops to ~30 by 1995, rises to ~45. Class C (dotted/circles): ~60 in 1975, drops to ~35 by 1995, drops to ~25 then ~30.]

To document the impact of prolonged drought on wetland ecosystems in the Murray-Darling Basin, researcher Kiera Thornton used satellite imagery to track changes in mean wetland area across habitat classifications. Sustained rainfall reductions in the 1980s and 1990s caused significant contraction in all wetland types. By 1995, mean wetland areas had dropped considerably. In that year, they were between _____.

- A) 100 and 150 hectares
- B) 60 and 80 hectares
- C) 30 and 45 hectares
- D) 140 and 180 hectares

Question 11 — Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Graph: A line graph titled “Insect Populations in a Welsh Meadow, March–December 2003” with months on x-axis, number observed (0–80) on y-axis. Meadow brown butterfly (solid/triangles): ~75 March, ~5 June, ~3 Sept, ~5 Dec. Common blue damselfly (dashed/squares): ~10 March, ~15 June, ~12 Sept, ~20 Dec. Hoverfly (dotted/circles): ~0 March, ~8 June, ~10 Sept, ~0 Dec.]

Eleri Griffiths and colleagues monitored insect populations in a meadow in Wales. They found that some species were entirely absent at particular times; for example, they did not observe even one _____.

- A) meadow brown butterfly in June of 2003.
- B) hoverfly in March and December of 2003.

- C) common blue damselfly in September of 2003.
- D) meadow brown butterfly in December of 2003.

Question 12 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Marine biologist Dr. Sayaka Ito examined otoliths of two reef fish species collected near Okinawa. Specimens labeled R-22, whose otolith growth rings formed during the juvenile stage, exhibited a chemical signature suggesting the fish had inhabited waters at 30 meters depth, while Okinawan reef fish of this species typically remain at depths of at least 50 meters. Because water temperature increases as depth decreases, a graduate student hypothesized that the Okinawan reef fish develop in warmer water than their counterparts near Palau do.

Which finding, if true, would most directly weaken the student's hypothesis?

- A) Okinawan reef fish tend to migrate to deeper water immediately after the juvenile stage, whereas Palauan reef fish do not.
- B) The water temperature at a given depth tends to be higher in waters near Palau than at that depth near Okinawa.
- C) There are not enough suitable shallow habitats near Palau for juvenile reef fish to develop at 30 meters.
- D) The water temperature at a given depth tends to be lower near Palau than at that depth near Okinawa.

Question 13 — Skill: Inference / Complete the Text | Difficulty: Hard

Numerous studies have identified a positive correlation between soil nitrogen content and plant species diversity in temperate grasslands across Western Europe. But Dr. Ingrid Halvorsen and Lars Ekman did not observe this correlation in Scandinavia, leading some ecologists to hypothesize that the pattern is restricted to Western Europe. However, Dr. Jun Tanaka reported nitrogen and diversity rising or falling together in Japan, and several other non-European studies yielded similar results, suggesting that _____.

- A) ecological characteristics unique to Scandinavia likely result in both metrics being significantly higher there than elsewhere.
- B) the Scandinavian grasslands may be more similar to Western European ones than to those in Japan.
- C) there may be circumstances specific to the Scandinavian grasslands that disrupted a general positive association between soil nitrogen and species diversity.
- D) grasslands outside Western Europe are more likely to show a negative association than a positive one.

Question 14 — Skill: Inference / Complete the Text | Difficulty: Hard

Dr. Meiling Chen et al. analyzed historical irrigation scaling trends circa 500 CE in East Asia, focusing on water management in ancient agricultural settlements. They examined communities across East Asia, with groupings based on regionally distinct political structures, crop types, and geographic constraints—including centrally administered rice-paddy systems of the Yangtze Delta, merchant-organized terrace farming in the Yunnan highlands, and village-managed canal systems in the Mekong lowlands. A prevalent trend emerged: as a settlement's total irrigated area increased, so did canal density in core agricultural zones. This finding implies that _____.

- A) even when there were substantial inter-regional differences in agricultural systems, these had a limited effect on irrigation patterns in ancient East Asian settlements.
- B) the observed relationship is heavily influenced by characteristics particular to ancient East Asia, making it unlikely a similar relationship would be found elsewhere.
- C) variations in political and crop-type interests had less effect within regions than between regions of East Asia.
- D) despite regional differences in political and agricultural systems, ancient East Asian settlements shared common geographic constraints that encouraged high canal density.

Question 15 — Skill: Grammar / Conventions | Difficulty: Medium

After extensive deliberation, _____ reached a consensus: the committee for the 1952 Pulitzer Prize in Fiction would award that year's prize to Herman Wouk for his sweeping novel set during World War II, praised for its vivid characterization and masterful narrative scope.

- A) anyone B) you C) they D) it

Question 16 — Skill: Grammar / Conventions | Difficulty: Medium

In September of 1971, the television program *All in the Family* was a cultural phenomenon. Having dominated the ratings for three consecutive months, _____ ranked No. 1 on the Nielsen chart of most-watched television shows.

- A) those B) they C) each D) it

Question 17 — Skill: Grammar / Conventions | Difficulty: Medium

In 1851, the novel *Moby-Dick* was published under the name Herman Melville. Today, we recognize that _____

- A) the author's ambition was to write an American epic,
 - B) the author's ambition was to write an American epic?
 - C) was the author's ambition to write an American epic.
 - D) the author's ambition was to write an American epic.
-

Question 18 — Skill: Grammar / Conventions | Difficulty: Medium

While the Iberian lynx (*Lynx pardinus*) can be found in scattered reserves across southern Spain and parts of Portugal in _____ than 90 percent of this critically endangered cat's population is concentrated in the Andalusia region. There, wildlife veterinarian Dr. Rosa Mendez leads a captive-breeding initiative.

- A) Europe, more B) Europe; more C) Europe. More D) Europe more
-

Question 19 — Skill: Grammar / Conventions | Difficulty: Medium

India and Mexico have two of the longest national anthems in the world. At 52 lines and 40 lines, respectively, these _____ according to research by musicologists Dr. Priya Sharma and David Cohen, correlate with periods of intense nationalist sentiment during which the anthems were composed.

- A) anthems' length, B) anthems lengths, C) anthem's length, D) anthems' lengths,
-

Question 20 — Skill: Grammar / Conventions | Difficulty: Medium

A lone figure gazes across a harbor in *Lighthouse at Two Lights*, an oil painting by American _____ Edward Hopper.

- A) artist: B) artist C) artist— D) artist,
-

Question 21 — Skill: Transitions | Difficulty: Medium

Every Canadian province has an associated official mineral, which is typically chosen by a panel of geologists and then ratified by the provincial legislature. For example, British Columbia's jade was formally designated in 1968, and Ontario's amethyst in 1975. _____ years elapse between a mineral's selection and its official recognition, as the legislative process can be remarkably slow.

- A) Often, B) Therefore, C) Similarly, D) Indeed,

Question 22 — Skill: Transitions | Difficulty: Medium

A 1995 study of flocking starlings observed the birds mimicking the flight direction of nearby flock-mates, rapidly banking away from an approaching hawk. This is an example of a social coordination mechanism that benefits the group. _____ the avian world is at times susceptible to the propagation of errors, often with costly outcomes: a 2012 study of migrating geese concluded that following erroneous course corrections by lead birds caused the flock to deviate from efficient routes unnecessarily.

- A) In other words, B) Conversely, C) Indeed, D) Consequently,
-

Question 23 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: *Nighthawks* is a 1942 oil-on-canvas painting by American artist Edward Hopper. It depicts four figures in a late-night diner. Oil painting is a technique in which pigments are mixed with a drying oil, typically linseed oil. The mixture is then applied to a prepared canvas or panel surface. Fresco painting is a different technique that applies pigments to wet plaster on a wall.

The student wants to describe the process of oil painting. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Oil painting uses linseed oil and other drying oils, not just wet plaster.
B) Hopper used oil-on-canvas, applying pigments mixed with oil onto canvas.
C) Hopper used oil painting to create the 1942 work *Nighthawks*.
D) In oil painting, pigments are combined with a drying oil and then applied onto a prepared canvas or panel.
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Question 24 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: Generally, a country's tallest building is located in its capital city. The capital of Malaysia is Kuala Lumpur. Malaysia's tallest building, the Merdeka 118 tower, is in Kuala Lumpur. The capital of Australia is Canberra. Australia's tallest building, however, is located in Melbourne, not Canberra.

The student wants to emphasize a difference between the two countries. Which choice most effectively uses relevant information from the notes?

- A) In both Malaysia and Australia, the tallest building is not in the capital city.
B) While Malaysia's tallest building is in its capital, Kuala Lumpur, Australia's tallest building is in Melbourne rather than its capital, Canberra.

- C) The tallest building in Malaysia, Merdeka 118, is in Kuala Lumpur, and Australia's capital is Canberra.
 - D) Malaysia's capital is Kuala Lumpur; Australia's capital is Canberra.
-

Question 25 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: Journalist Sonia Nazario's book *Enrique's Journey* is about child migration from Central America. Central American child migration surged in the early 2000s with tens of thousands of unaccompanied minors. To document this, Nazario retraces the journey of a Honduran teenager named Enrique, among others. The book won the Grand Prize of the Robert F. Kennedy Book Award in 2007. It was also a Pulitzer Prize finalist in 2007.

Which choice most effectively uses information to emphasize how Nazario documents child migration?

- A) Honduras is one of the countries from which children migrated to the United States.
 - B) Sonia Nazario chronicles child migration in personal detail by retracing the journey of individuals, such as the Honduran teenager Enrique.
 - C) In 2007, Nazario's book won the Robert F. Kennedy Book Award and was a Pulitzer finalist.
 - D) The journey of Enrique is featured in the award-winning book *Enrique's Journey*.
-

Question 26 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: Yo-Yo Ma is a French-American cellist and composer. His acclaimed albums feature many distinguished guest artists. Pianist Kathryn Stott performed on his 2015 album, *Songs of Comfort and Hope*. Guitarist Carlos Santana performed on his 2008 album, *Songs of Joy and Peace*.

The student wants to emphasize a similarity between Kathryn Stott and Carlos Santana. Which choice most effectively uses the notes?

- A) Stott performed on Ma's 2015 album, released seven years after *Songs of Joy and Peace*.
 - B) *Songs of Comfort and Hope* and *Songs of Joy and Peace* are albums by Yo-Yo Ma.
 - C) Although both performed on Ma's albums, Stott played piano whereas Santana played guitar.
 - D) Kathryn Stott and Carlos Santana both contributed their musical talents to albums by Yo-Yo Ma.
-

Question 27 — *Skill: Rhetorical Synthesis | Difficulty: Medium*

While researching a topic, a student has taken the following notes: Roberto Clemente (1934–1972) was a baseball player born in Carolina, Puerto Rico. He was the first Caribbean-born player inducted into the Baseball Hall of Fame. Early in his career, he played in limited, low-profile minor league games, such as his stint with the Montreal Royals (1954). Later, he became one of the most celebrated outfielders in Major League history with the Pittsburgh Pirates (1955–1972).

The student wants to make and support a generalization about Clemente’s early career. Which choice most effectively uses the notes?

- A) His stint with the Montreal Royals (1954) came early, whereas Pittsburgh came later.
- B) Roberto Clemente’s earliest professional appearances were modest and low-profile, such as his stint with the Montreal Royals in 1954, where he played in limited games.
- C) Both the Montreal Royals and the Pittsburgh Pirates are teams that include Clemente.
- D) With the Montreal Royals in 1954, Clemente played in limited minor league games.

SECTION 1: READING AND WRITING

MODULE 2 — EASIER PATH (27 Questions)

For students scoring below ~60% on Module 1 — ~70% Easy/Medium, ~30% Hard

Question 1 — Skill: Words in Context | Difficulty: Easy

Despite vocal support from environmental groups, the proposed marine conservation act gained little traction in the legislature during the 2010s, largely because lawmakers lacked data from developing coastal nations. Advocates would later _____ this gap by partnering with researchers in Southeast Asia and East Africa.

- A) reinforce B) assume C) classify D) address
-

Question 2 — Skill: Words in Context | Difficulty: Easy

Although the metabolic processes of deep-sea bacteria have received limited formal investigation, indirect measurements have suggested that these organisms generate internal temperatures above their surroundings. Laboratory findings by Dr. Akira Sato et al. now _____ this idea: cultured colonies maintained elevated metabolic heat relative to incubation chambers.

- A) contradict B) confirm C) summarize D) limit
-

Question 3 — Skill: Words in Context | Difficulty: Easy

The pacing of Kurosawa’s epic films is notably measured and contemplative. Georgian filmmaker Otar Iosseliani has acknowledged that his meditative works may lull viewers to sleep. This might imply tedium, but slow pacing does not necessarily produce boredom: deliberate rhythm generates focused alertness, and thus induces not torpor but _____.

- A) resilience B) agitation C) serenity D) attentiveness
-

Question 4 — Skill: Words in Context | Difficulty: Easy

Challenges in recording nocturnal amphibian calls and a prevailing emphasis on diurnal species have resulted in _____ of data on nighttime vocalization patterns, but Dr. Elena Vasquez has begun to close this gap by deploying automated acoustic monitors.

- A) a surplus B) a collection C) a scarcity D) a misrepresentation
-

Question 5 — Skill: Overall Structure | Difficulty: Easy

Vertical inheritance involves the transfer of cultural knowledge from elder to younger generation; horizontal transmission involves sharing between contemporaries. While vertical transmission is common in apprenticeship-based crafts, it has rarely been documented among peer-learning communities. However, new research suggests that horizontal cultural

Which choice best states the function of the underlined sentence in the text as a whole?

- A) It implies that a common understanding of horizontal cultural transmission may be inaccurate.
- B) It compares the rates at which horizontal transmission has been observed in two types of communities.
- C) It argues that a particular line of research is unlikely to yield results.
- D) It distinguishes between horizontal and vertical cultural transmission.

Question 6 — Skill: Main Purpose | Difficulty: Medium

Middlemarch, first published serially in 1871–1872, is a novel in George Eliot’s body of work. Some critics have found fault with its expansive scope, saying it meanders and loses narrative focus. But other critics argue the novel should not be thought of as loosely structured but as an intricately woven tapestry, similar to Balzac’s *La Comédie humaine*.

Which choice best states the main purpose of the text?

- A) To present a reason the length of *Middlemarch* makes it challenging
- B) To argue that Eliot’s novels should have the reputation of *La Comédie humaine*
- C) To describe a characteristic of Eliot’s novel and offer two differing viewpoints on it
- D) To explain why critics find Eliot’s novels entertaining despite structural flaws

Question 7 — Skill: Overall Structure | Difficulty: Medium

With its blend of blues and country influences, Folsom Prison Blues is representative of Johnny Cash’s recordings in the mid-1950s and early 1960s. Through songs straddling these genres, he achieved his greatest early recognition. A decade earlier, however, Cash had first honed his skills as a gospel singer in rural Arkansas. The imprint of this sacred music tradition can be detected in I Walk the Line.

- A) It explores Cash’s lifelong interest in gospel music, then debates whether country music influenced his work.
- B) It discusses a particular period in Cash’s career, then considers the music of an earlier period.

- C) It enthusiastically recommends Cash's earliest recordings, then expresses disappointment with later ones.
- D) It praises one song, then notes how similar it is to a song Cash recorded earlier.

Question 8 — Skill: Cross-Text | Difficulty: Medium

Text 1: Research on psychological benefits of exercise has traditionally concentrated on structured workouts, but kinesiologist Dr. Amara Osei has demonstrated that incidental physical activity—such as walking to a bus stop—also contributes measurably to improved mood. Greater positive affect was reported by participants who incorporated more incidental movement.

Text 2: Exercise psychology commonly draws on a framework categorizing physical activity along three intensity levels: sedentary (watching TV), light (casual walking), and vigorous (competitive running).

Based on the texts, what would Osei (Text 1) most likely say about the framework in Text 2?

- A) It highlights that most research overlooks a category of physical activity that can improve emotional well-being.
- B) It reflects an overemphasis on vigorous activity in evaluations of exercise importance.
- C) It explains observations that individuals expect structured workouts to be more mood-enhancing than incidental movements.
- D) It emphasizes distinctions among types of vigorous activity not adequately represented in exercise research.

Question 9 — Skill: Cross-Text | Difficulty: Easy

Text 1: From the ancient giant wombat to the living koala, marsupials are among Australia's most distinctive mammals. But zoologists still have much to learn. Today's marsupials spend much of their early development in pouches. The pouch's inaccessibility has made it difficult to study neonatal development.

Text 2: By using miniature thermal sensors, Dr. Hannah Liu can now monitor temperature fluctuations within marsupial pouches. Such sensors can revise misconceptions. It was assumed pouch temperatures were constant. But pouch temperatures in wild kangaroos fluctuate by up to °C over 24 hours.

The author of Text 1 and Text 2 both discuss which topic?

- A) How extinct marsupials compare with living marsupials
- B) Scientists' efforts to understand marsupials
- C) The benefits of attaching monitoring devices to marsupials

D) Changes in marsupial habitats

Question 10 — Skill: Inference / Complete the Text | Difficulty: Medium

Humpback whales consume up to 1.5% of their body mass in krill per day. So a 30,000 kg humpback could eat 450 kg, while a 150,000 kg blue whale could consume 2,250 kg. Over the past century, baleen whale populations have declined, and—contrary to expectations—so have krill populations. Dr. Smetacek explains that great whales recycle iron through fecal plumes, fertilizing phytoplankton blooms that sustain krill.

Based on the text, what can most reasonably be concluded about humpback and blue whales?

- A) As humpback whale krill consumption has decreased, blue whale krill consumption has increased.
- B) The blue whale is able to consume more krill per day than the humpback whale.
- C) Both can eat krill, but only the humpback can also eat small fish.
- D) Both populations have declined due to decreasing ocean iron levels.

Question 11 — Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Graph: Grouped bar chart showing correlation between algorithm-predicted and user-reported satisfaction ratings for Classical and Electronic music. Three users: U5, U1, U2. For Classical: U5~0.25, U1~0.38, U2~0.55. For Electronic: U5~0.27, U1~0.22, U2~0.25.]

Dr. Nakamura developed a model to predict song enjoyment on a 1–4 scale, then recruited participants to rate songs. The team computed correlation between predicted and reported ratings. Assuming U1 and U2 gave equal ratings to classical tracks, the graph indicates the model predicted _____.

- A) U1's and U2's classical ratings would equal one another.
- B) U1's and U2's classical ratings would differ from one another.
- C) U1 would derive more satisfaction from classical than from electronic tracks.
- D) U1 would derive less satisfaction from classical than from electronic tracks.

Question 12 — Skill: Quotation Illustrates Claim | Difficulty: Hard

A Doll's House is an 1879 play by Henrik Ibsen. In the play, Torvald Helmer has been promoted at the bank, and his wife Nora conceals the fact that she once forged a signature to save his life. Ibsen portrays Nora's tension as increasingly unbearable, as is evident when _____.

Which quotation from a translation most effectively illustrates the claim?

- A) Torvald says, “Is it my little squirrel bustling about? Don’t disturb me.”
- B) Nora says to herself, “Corrupt my own children? Poison my home? It can never, never be true.”
- C) Torvald says of a colleague, “He is morally lost, and yet he goes on living.”
- D) Dr. Rank tells Nora, “There is one last thing I should like to do.”

Question 13 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Certain industrial processes release TiO_2 nanoparticles into waterways. Dr. Bergström found TiO_2 -NPs accumulate in earthworms. It has been hypothesized that TiO_2 -NP monitoring in earthworms could serve as a proxy, replacing costly sampling of vertebrate species like European moles, commonly used in regulatory soil-quality testing.

Which finding, if true, would most directly weaken the hypothesis?

- A) Compared with moles, earthworms can accumulate TiO_2 -NPs with significantly fewer negative health effects.
- B) The rate of TiO_2 -NP uptake in earthworms differs from that in moles in a way not yet understood.
- C) Earthworms accumulate several other nanoparticle types, whereas moles accumulate only TiO_2 -NPs.
- D) When both species are exposed to similar TiO_2 -NP levels, concentrations show little variation.

Question 14 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. Alvarez investigated bioluminescence in deep-sea squid during simulated predator encounters. 40% of tested species produced light in three patterns: steady glows, rapid flashes, and pulsing waves. Two species released bioluminescent ink and exclusively produced rapid flashes. Alvarez argues bioluminescence in ink-releasing species functions primarily to startle predators.

Which finding, if true, would most directly support Alvarez’s claim?

- A) Brief, intense light bursts like rapid flashes tend to be especially effective in triggering a startle reflex in deep-sea predators.
- B) Ink-producing squid generate bioluminescence at a higher intensity than non-ink-producing squid.
- C) Rapid flashes are particularly visible to sharks and large fish.
- D) Sperm whales halt attacks in response to bioluminescent signals of any duration.

Question 15 — Skill: Inference / Complete the Text | Difficulty: Medium

Archaeologists assume that population displacement interrupts intergenerational craft transmission, causing simultaneous declines in multiple crafting traditions. Inventories from the Indus Valley show a steep decline around 1900 BCE in carnelian bead-drilling, an advanced technique. But advanced bronze-casting and most other traditions continued to flourish, suggesting that _____.

- A) disruptions in bead-drilling likely occurred in the context of demographic continuity in the Indus Valley.
- B) a simultaneous decline in multiple traditions is less likely to indicate disrupted expertise transmission.
- C) peoples continued to uphold multiple traditions even as population shifts occurred.
- D) cross-cultural exchange likely explains the expansion of bronze-casting beginning around 1900 BCE.

Question 16 — Skill: Grammar / Conventions | Difficulty: Medium

In a chapter from Matsuo Basho's *The Narrow Road to the Deep North*, the author marvels at a polished floor, wind chimes, and a child's laughter. Such observations, recorded by Basho as he journeyed through seventeenth-century Japan, _____ the ephemeral beauty of everyday moments.

- A) illuminates B) was illuminating C) to illuminate D) illuminate

Question 17 — Skill: Grammar / Conventions | Difficulty: Easy

Citing the example of Athena, the philosopher Porphyry theorized that all Greek deities—including Poseidon, the god of the sea; Hermes, the god of trade; and the god of the forge, _____ may have been historical rulers who underwent deification.

- A) Hephaestus; B) Hephaestus— C) Hephaestus, D) Hephaestus

Question 18 — Skill: Grammar / Conventions | Difficulty: Medium

Classified as a spondee, the word "sunlight" consists of two equally stressed syllables. The word "guitarist," by contrast, consists of one unstressed syllable followed by two stressed syllables, a metrical foot _____ to in English prosody as an anapest.

- A) referred B) refers C) referring D) is referred

Question 19 — Skill: Grammar / Conventions | Difficulty: Medium

The libraries of Portland, Oregon, seem to foster greater civic engagement. In a 2021 study, researchers investigating the connection between library branch locations and voter participation _____ analyzed geotagged voter registration data from various neighborhoods. They found neighborhoods with library branches had higher voter registration rates.

- A) conducted and the study's scope
- B) conducted, and the study's scope
- C) conducted and the study's scope,
- D) conducted, and the study's scope,

Question 20 — Skill: Grammar / Conventions | Difficulty: Medium

With a generating capacity of 48 megawatts, the Hellisheidi geothermal plant in Iceland provides reliable renewable energy. Plants like Hellisheidi supply a substantial fraction of Iceland's _____ the rarity of locations with the required volcanic activity and permeable rock limiting broader adoption.

- A) electricity; though, B) electricity, though; C) electricity, though D) electricity, though,

Question 21 — Skill: Transitions | Difficulty: Easy

The album *Buena Vista Social Club* received widespread praise for its celebration of son cubano, a genre featuring layered harmonies and guitar-driven rhythms. _____ few were surprised when it won the Grammy for Best Tropical Latin Album in 1998.

- A) Often, B) Granted, C) Nevertheless, D) Therefore,

Question 22 — Skill: Transitions | Difficulty: Easy

In 1925, the Coast and Geodetic Survey calculated Montana's geographic center using only a plaster model and a balancing pin. _____ this seemingly imprecise method yielded a remarkably accurate result: the 1925 center diverged from a satellite-computed center by just a few kilometers.

- A) Still, B) Similarly, C) In other words, D) Fittingly,

Question 23 — Skill: Transitions | Difficulty: Medium

In an ekphrastic novel, a writer engages with a visual artwork. Tracy Chevalier's *Girl with a Pearl Earring* imagines the story behind Vermeer's 1665 painting; Michael Cunningham's *Specimen Days*, _____ takes as its touchstone a Walt Whitman poem rather than a visual artwork.

- A) nevertheless, B) hence, C) likewise, D) conversely,
-

Question 24 — Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes: Choreographers design and arrange dance movements. They work with dancers to translate artistic visions. *Swan Lake* (1877) was choreographed by Julius Reisinger. Twyla Tharp choreographed dance sequences for productions.

The student wants to provide an example of a production a choreographer worked on. Which choice is most effective?

- A) Choreographer Twyla Tharp worked with dancers to translate artistic visions.
B) The 1877 production of *Swan Lake* is one example of Julius Reisinger's work as a choreographer.
C) In dance, choreographers design and arrange movements.
D) Twyla Tharp works with dancers to translate a director's vision into movement.
-

Question 25 — Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- The Mandarin word yuan comes from a Chinese character meaning "round."
- Multiple East Asian currencies use some form of this word.
- China's currency is the renminbi, with the basic unit called the yuan.

The student wants to provide an example of a country using "yuan" in its currency name. Which choice is most effective?

- A) The Chinese character meaning "round" is the origin of "yuan."
B) Several countries use a form of "yuan" in naming their currencies.
C) China uses the yuan as the basic unit of its national currency, the renminbi.
D) China does not refer to its currency by the original Chinese character.
-

Question 26 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: The ratio of an element's mass to the combined mass of all elements is the mass fraction. Silicon's mass fraction in Earth's crust is 277,000 ppm (27.7%). The mole fraction is the ratio of atom count. Silicon's mole fraction is ~20.5%.

The student wants to emphasize a similarity between mass fraction and mole fraction. Which choice is most effective?

- A) Silicon's mass fraction is 277,000 ppm, or 27.7%, and its mole fraction is about 20.5%.
- B) While mass fraction measures relative mass and mole fraction measures relative atom count, both indicate relative abundance of elements.
- C) Both express silicon's abundance as a ratio to the total.
- D) The mass fraction indicates relative mass; the mole fraction indicates relative atom number.

Question 27 — Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: Architect Zaha Hadid (1950–2016) was born in Baghdad. She was the first woman to receive the Pritzker Architecture Prize. Early in her career, she produced many unbuilt designs, such as her Peak competition entry (Hong Kong, 1983). Later, she realized iconic structures like the Heydar Aliyev Center (Baku, 2012).

The student wants to generalize about Hadid's early career. Which choice is most effective?

- A) Her Peak entry (1983) came early; the Heydar Aliyev Center came later.
- B) Hadid's earliest works were largely theoretical and unrealized, such as her visionary 1983 Peak competition entry, which was never constructed.
- C) Both the Peak entry and Heydar Aliyev Center are by Hadid, the first woman to win the Pritzker Prize.
- D) For the Peak competition in Hong Kong in 1983, Hadid produced an unbuilt design.

SECTION 1: READING AND WRITING

MODULE 2 — HARDER PATH (27 Questions)

For students scoring above ~60% on Module 1 — ~30% Easy/Medium, ~70% Hard

Question 1 — Skill: Words in Context | Difficulty: Hard

Although conventional wisdom held that Ottoman mosque design was monolithically derived from Byzantine precedents, recent scholarship by Dr. Fatma "Ozkan has demonstrated that provincial Ottoman architects in the Balkans frequently _____ local Slavic building traditions, incorporating steeply pitched timber roofing and asymmetrical courtyard plans.

- A) repudiated B) supplanted C) catalogued D) assimilated

Question 2 — Skill: Words in Context | Difficulty: Hard

The notion that mycorrhizal networks facilitate directional nutrient transfer from established trees to struggling seedlings has achieved wide currency in popular science. However, Dr. Justine Karst argues that empirical support for this claim remains _____: most studies have been conducted under controlled conditions that poorly replicate natural forest conditions.

- A) unambiguous B) voluminous C) irrelevant D) tenuous

Question 3 — Skill: Main Purpose | Difficulty: Hard

The compositional approach of *Migrations*, a 2020 orchestral work by Cherokee-Creek composer Jerod Tate, aligns more closely with post-minimalist concert music than with Southeastern Woodlands Indigenous music. Few listeners would deduce from its sonic texture that its composer is Indigenous. In contrast, Diné composer Connor Chee builds compositions from Navajo chant melodies. Thus in Chee's work, classical form has the effect of _____.

- A) rendering Indigenous musical identity more legible than it is in *Migrations*.
B) asserting the Indigenous origins of motifs associated with post-minimalism.
C) challenging the dominance of European American composers.
D) reconciling Indigenous and European American influences that coexist uneasily in *Migrations*.
-

Question 4 — Skill: Inference / Complete the Text | Difficulty: Hard

Historians of technology assume that geopolitical disruption interrupts technical expertise circulation, manifesting as simultaneous patent complexity reductions. Rhineland patent archives from 1910–1950 show a steep drop around 1918 in precision optical grinding patents. But advanced chemical synthesis patents continued at pre-war rates, suggesting that _____.

- A) disruptions in optical patents likely occurred amid continuity in the broader Rhineland engineering workforce.
- B) a simultaneous decline across multiple engineering domains is less likely to indicate disrupted expertise circulation.
- C) engineers continued to uphold multiple traditions even as geopolitical disruptions occurred.
- D) cross-border exchange likely explains the expansion of chemical synthesis beginning around 1918.

Question 5 — Skill: Inference / Complete the Text | Difficulty: Hard

Leafcutter ants cultivate fungal gardens for food. When gardens become contaminated with the parasitic mold *Escovopsis*, worker ants selectively remove infected portions, even though this temporarily reduces food supply. Researchers argue this weeding behavior, while costly short-term, is adaptive because _____.

- A) the colony can compensate by foraging for alternative vegetation.
- B) workers consume *Escovopsis* directly as an alternative protein source.
- C) an unchecked infection would destroy the entire garden, permanently eliminating the colony's sole food source.
- D) discarded material can be recycled by other soil insect species.

Question 6 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. Ramaswamy observed that congestion pricing in European cities coincided with increased public transit ridership. She hypothesizes that congestion pricing directly causes commuters to switch to transit. However, the cities studied had simultaneously invested in substantial transit improvements.

Which finding, if true, would most directly weaken her hypothesis?

- A) In cities with comparable transit improvements but no congestion pricing, ridership increased by a similar magnitude.
- B) The congestion pricing fees were set at moderate levels.
- C) Some commuters reported switching to transit primarily because of the fees.

- D) The cities experienced a slight overall decrease in total commuting trips.

Question 7 — Skill: Cross-Text | Difficulty: Hard

Text 1: Dr. Kaplan found that 20 minutes in green spaces significantly improved attention restoration compared to urban settings. Natural environments engage involuntary attention, allowing directed-attention mechanisms to recover.

Text 2: Dr. Joye argues that benefits attributed to nature may partially stem from reduced sensory complexity and lower noise rather than intrinsic restorative properties. Carefully designed quiet urban spaces can produce comparable attention benefits.

How would Joye (Text 2) most likely respond to the findings in Text 1?

- A) The improvements may not be attributable specifically to the natural character of the environments studied.
- B) Urban environments are inherently more cognitively demanding.
- C) Twenty minutes is insufficient for meaningful attention restoration.
- D) Directed-attention mechanisms cannot be measured reliably in field studies.

Question 8 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. Nordqvist proposes that minority language decline in urban areas is driven not by discrimination but by “prestige asymmetry”—the unconscious association of a dominant language with economic opportunity. Even in cities with legal protections, speakers may abandon their heritage language in professional contexts.

Which finding, if true, would most directly support Nordqvist’s claim?

- A) In cities with strong legal protections, bilingual speakers increasingly use the dominant language at work while maintaining the minority language only at home.
- B) Cities with minority language education programs see a temporary increase in fluent speakers.
- C) The rate of decline is roughly equal with and without legal protections.
- D) Rural speakers maintain their heritage language at higher rates than urban speakers.

Question 9 — Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Graph: Scatterplot with x (0–50) and y (0–70). Line of best fit has negative slope, y -intercept ~ 58 .] Data set Q is created by adding 7 units to y for each data point. What is closest to the y -intercept of the line of best fit for data set Q?

- A) 51 B) 58 C) 65 D) 72
-

Question 10 — Skill: Grammar / Conventions | Difficulty: Hard

Niels Bohr once remarked that predictions are difficult—especially about the future. This observation, while humorous, captures a genuine epistemological _____ the limits of scientific forecasting become most apparent precisely when the stakes of accurate prediction are highest.

- A) tension, B) tension— C) tension; D) tension:
-

Question 11 — Skill: Grammar / Conventions | Difficulty: Hard

The discovery of high arsenic concentrations in Bangladesh wells, initially documented by Dr. Ravenscroft in the 1990s, alongside subsequent epidemiological studies linking arsenic exposure to elevated cancer rates, _____ prompted the WHO to revise recommended maximum contaminant levels.

- A) have B) has C) are D) were
-

Question 12 — Skill: Transitions | Difficulty: Hard

Dr. Diallo’s research shows that post-conflict communities develop shared narratives that simplify complex causes into morally clear stories. _____ these simplified narratives, while psychologically comforting, can impede reconciliation by obscuring ambiguous roles individuals played during the conflict.

- A) Similarly, B) For instance, C) However, D) In other words,
-

Question 13 — Skill: Inference / Complete the Text | Difficulty: Hard

The concept of “critical periods”—windows when the brain is especially receptive to specific input—has been foundational in neuroscience. However, Dr. Hensch has revealed that pharmacological interventions can reopen plasticity in the adult visual cortex of mice. This finding suggests that _____.

- A) the neural mechanisms governing critical periods are more modifiable than the traditional model implies, potentially opening therapeutic avenues for conditions considered untreatable after childhood.
- B) critical periods in cats and mice are governed by fundamentally different mechanisms.

- C) pharmacological interventions are more effective than sensory experience in shaping visual cortex development.
 - D) the concept of critical periods should be abandoned entirely.
-

Question 14 — Skill: Transitions | Difficulty: Hard

Dr. Sjöberg _____ has demonstrated that consumers' willingness to pay a premium for ethically sourced products declines when they also see environmental impact information. _____ a consumer who would pay 20% more for fair-trade coffee might pay only 8% more if the label also discloses the product's carbon footprint.

- A) Nevertheless,
 - B) Subsequently,
 - C) In contrast,
 - D) For example,
-

Question 15 — Skill: Grammar / Conventions | Difficulty: Hard

The architectural innovations of three Renaissance _____ palaces demonstrate distinct approaches to the relationship between interior light and structural form.

- A) architect's
 - B) architects
 - C) architects'
 - D) architects's
-

Question 16 — Skill: Inference / Complete the Text | Difficulty: Hard

Paleoclimatologists have used oxygen isotope ratios in ice cores as a temperature proxy. But Dr. Gkinis showed that during abrupt climate transitions, the ratio can be influenced by changes in atmospheric moisture source region, not just temperature. This implies that _____.

- A) isotope-based temperature reconstructions for rapid climate change periods may be less reliable than those for stable periods.
 - B) the isotope ratio is more influenced by moisture source changes than by temperature in all layers.
 - C) Dansgaard-Oeschger events were likely less abrupt than isotope data suggest.
 - D) alternative proxies are uniformly more accurate for all time periods.
-

Question 17 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. McAllister hypothesizes that the increase in peregrine falcon populations in North American cities is primarily attributable to the abundance of rock pigeons as prey. She notes that urban peregrines prey almost exclusively on pigeons, which have remained abundant.

Which finding, if true, would most directly weaken this hypothesis?

- A) Peregrine populations have increased at comparable rates in rural areas where pigeons are absent but similar-sized prey is abundant.
- B) Rock pigeon populations in some cities have actually declined slightly.
- C) Urban peregrines occasionally prey on starlings and sparrows.
- D) City buildings provide nesting ledges resembling natural cliff faces.

Question 18 — Skill: Rhetorical Synthesis | Difficulty: Hard

Notes:

- Architect Kere was born in Gando, Burkina Faso, in 1965.
- He was the first African to win the Pritzker Prize (2022).
- His philosophy emphasizes community participation and local materials.
- For Gando Primary School (2001), villagers contributed labor and used compressed laterite clay bricks made on-site.

The student wants to illustrate how Kere's philosophy manifests in a specific project. Which choice is most effective?

- A) Kere was the first African architect to receive the Pritzker Prize.
- B) Kere's philosophy emphasizes community participation and local materials.
- C) The Gando Primary School (2001) was Kere's first major project.
- D) Kere's Gando Primary School exemplifies his philosophy: villagers made compressed laterite clay bricks on-site and contributed their labor.

Question 19 — Skill: Rhetorical Synthesis | Difficulty: Hard

Notes: The Drake Equation estimates communicative civilizations in the Milky Way. Formulated by Frank Drake in 1961. Critics argue several factors are essentially unknown. Supporters say it provides a useful framework for scientific inquiry.

The student wants to present the debate about the Drake Equation's usefulness. Which choice is most effective?

- A) The Drake Equation multiplies seven factors to estimate communicative civilizations.
- B) While critics argue key factors remain essentially unknown, supporters maintain the equation serves as a valuable framework for structuring scientific investigation.
- C) Frank Drake formulated his equation in 1961, and it has been both criticized and supported.
- D) The equation includes the rate of star formation and the fraction of planets developing life.

Question 20 — Skill: Grammar / Conventions | Difficulty: Hard

Dr. Huang's longitudinal study found that code-switching does not reflect linguistic _____ it instead demonstrates sophisticated metalinguistic awareness, as code-switching children show enhanced executive function compared to monolingual peers.

- A) confusion, B) confusion. C) confusion D) confusion;
-

Question 21 — Skill: Grammar / Conventions | Difficulty: Hard

The collaborative research initiative between marine biologists at Woods Hole and glaciologists at the University of Iceland, which began in 2018 and _____ annual expeditions to monitor Vatnajökull's marine-terminating glaciers, has produced the most comprehensive dataset on subglacial meltwater discharge.

- A) has involved B) involving C) involve D) to involve
-

Question 22 — Skill: Transitions | Difficulty: Hard

Dr. Patel demonstrated that listeners' emotional responses to unfamiliar non-Western musical scales are more nuanced than assumed. Rather than simply rating them as "dissonant," participants identified distinct emotional qualities—melancholy, tension, serenity. _____ these findings challenge the view that emotional responses to music are primarily shaped by cultural familiarity.

- A) Similarly, B) For instance, C) Taken together, D) In other words,
-

Question 23 — Skill: Grammar / Conventions | Difficulty: Hard

Discovered by amateur fossil hunter Mary Anning on the Dorset coast in 1811, _____ the ichthyosaur skeleton challenged prevailing scientific assumptions about the history of life on Earth.

- A) scientists were forced to reconsider extinction—
B) the discovery was remarkable in its implications:
C) it was a remarkable achievement, and
D) the significance of the find was immediately recognized:
-

Question 24 — Skill: Inference / Complete the Text | Difficulty: Hard

Dr. Asante documented that the colonial transition from oral to written legal codes in West Africa did not result in wholesale replacement of customary law. Instead, colonial courts operated a dual system: written statutes governed commercial disputes while customary traditions adjudicated family law and land inheritance. This persistence suggests that _____.

- A) the imposition of written codes was more selective in its effects than a simple narrative of cultural displacement would imply, with customary practices proving resilient in domains central to community social structure.
- B) colonial administrators deliberately preserved customary law to avoid provoking resistance.
- C) written codes were adopted more rapidly in urban centers than in rural areas.
- D) customary oral traditions were more formally structured than colonial scholars recognized.

Question 25 — Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. Zakharova proposes that high metallicity in ancient globular clusters can be explained by “sequential self-enrichment”: first-generation massive stars explode as supernovae, seeding remaining gas with heavier elements before a second generation forms. She argues this occurs within a single cluster-formation episode.

Which finding would most directly support her claim?

- A) Spectroscopic analysis reveals two distinct chemical abundance patterns within the same cluster, consistent with successive generations from progressively enriched gas.
- B) Simulations show forming clusters cannot retain supernova ejecta.
- C) High-metallicity clusters are preferentially located near galactic centers.
- D) Age spread within most clusters is less than 100 million years.

Question 26 — Skill: Rhetorical Synthesis | Difficulty: Hard

Notes: The Antikythera mechanism was recovered from a shipwreck in 1901. Dates to ~100 BCE. Contains at least 30 interlocking bronze gears. Predicted eclipses and tracked planetary positions. No comparably complex device existed for another 1,000 years.

The student wants to emphasize the historical significance of the mechanism’s technical sophistication. Which choice is most effective?

- A) The mechanism dates to approximately 100 BCE and was recovered in 1901.

- B) With at least 30 interlocking bronze gears capable of predicting eclipses and tracking planetary positions, the Antikythera mechanism represents a level of complexity not seen again for a millennium.
- C) The ancient Greeks knew of five planets, and the mechanism could track all of them.
- D) Researchers have identified at least 30 gears within the mechanism.

Question 27 — Skill: Transitions | Difficulty: Hard

Voyager 1 continues to transmit data from interstellar space more than four decades after launch. Its instruments, designed for a five-year mission, have far outlasted their projected lifespan. _____ the plutonium-238 generators are gradually losing output, and NASA engineers estimate that by the mid-2030s, insufficient power will remain.

- A) Furthermore, B) In other words, C) For example, D) Nevertheless,

SECTION 2: MATH

MODULE 1 (Same for all students — 22 Questions)

Question 1 — Skill: Linear Equations in One & Two Variables | Difficulty: Easy

[Figure: A quadrilateral with two sides of length a cm and two sides of length b cm.] The perimeter of the quadrilateral is 58 centimeters. Which equation represents this situation?

- A) $a + b = 58$ B) $2a + 2b = 58$ C) $a + 2b = 58$ D) $2a + b = 58$
-

Question 2 — Skill: Function Notation & Evaluation | Difficulty: Easy

If $5 + x = 3$, what is the value of $30 + 6x$?

Question 3 — Skill: Creating Linear Models from Word Problems | Difficulty: Easy

A trail mix recipe calls for 3 cups of almonds and 5 bags of dried cranberries and has 1,740 total calories. There are 160 calories per cup of almonds. How many calories are in each bag of dried cranberries?

- A) 1,260 B) 540 C) 252 D) 108
-

Question 4 — Skill: Percentages | Difficulty: Easy

A bookstore offers hardcover books at a 15% discount. What is the discount, in dollars, on a book with an original price of \$180?

- A) 153 B) 108 C) 45 D) 27
-

Question 5 — Skill: Ratios, Rates & Proportions | Difficulty: Easy

A recipe for 8 servings of soup requires 3 cups of broth. At this rate, how many cups of broth are needed for 24 servings?

- A) 6 B) 9 C) 11 D) 16
-

Question 6 — Skill: Systems of Linear Equations | Difficulty: Medium

$5x = 20$; $-2x + y = -3$. The solution is (x, y) . What is the value of $x + y$?

- A) -9 B) -1 C) 9 D) 11
-

Question 7 — Skill: Linear Inequalities | Difficulty: Medium

A customer buys notebooks at \$1.25 each with a \$3 store credit. No sales tax. Which inequality represents the number of notebooks, n , purchasable for no more than \$15?

- A) $1.25n + 3 \leq 15$ B) $1.25n + 3 \geq 15$ C) $1.25n - 3 \leq 15$ D) $1.25n - 3 \geq 15$
-

Question 8 — Skill: Circle Equations | Difficulty: Easy

Circle W has a radius of 24 cm. What is the area of circle W , in cm^2 ?

- A) 24π B) 48π C) 288π D) 576π
-

Question 9 — Skill: Circle Equations | Difficulty: Medium

Circle M : $(x + 5)^2 + (y + 3)^2 = 25$. Circle N has the same center but a radius 3 units greater. N is defined by $(x + 5)^2 + (y + 3)^2 = p$. What is p ?

Question 10 — Skill: Polynomial Operations | Difficulty: Easy

Which expression is equivalent to $5ab^2 + 3ab + 2ab^2$?

- A) $7ab^2 + 3ab$ B) $7ab^3 + 3ab$ C) $10ab^3$ D) $10ab^5$
-

Question 11 — Skill: Ratios, Rates & Proportions | Difficulty: Medium

If $\frac{3a}{2b} = 84$, what is the value of $\frac{a}{b}$?

Question 12 — Skill: Similar Triangles | Difficulty: Medium

[Figure: Two similar triangles with angle $PST = 63^\circ$ and $QR = 15(ST)$.] What is the measure of angle PQR ?

- A) $(15 \times 63)^\circ$ B) $(15 + 63)^\circ$ C) 63° D) 15°

Question 13 — Skill: Graphs of Linear Equations | Difficulty: Medium

[Graph: Curve showing estimated melting point y °C vs carbon atoms x in a straight-chain alcohol, passing through (12.5, 24.8).] Which is the best interpretation of the point (12.5, 24.8)?

- A) An alcohol with 24.8 carbons has a melting point of 12. °C
- B) An alcohol with 12.5 carbons has a melting point of 24. °C
- C) The minimum melting point corresponds to 12.5 carbons.
- D) The maximum melting point corresponds to 12.5 carbons.

Question 14 — Skill: Graphs of Linear Equations | Difficulty: Medium

A linear function g has $g(2) = 10$ and $g(5) = 22$. What is the value of $g(0)$?

- A) 2 B) 4 C) 6 D) 8

Question 15 — Skill: Linear Equations in One & Two Variables | Difficulty: Medium

If $\frac{90}{x} = 5x$, what is the value of $5x^2$?

- A) 450 B) 5 C) 18 D) 90

Question 16 — Skill: Linear Inequalities | Difficulty: Medium

$y \leq 3x - 14$. Which point (x, y) is a solution?

- A) $(-3, -14)$ B) $(0, 14)$ C) $(7, 7)$ D) $(4, 14)$

Question 17 — Skill: Linear Inequalities | Difficulty: Medium

A gardener plants at most 12 bulbs total (tulips x and daffodils y). Tulips need 2 sq in, daffodils need 3 sq in, with 30 sq in available. Which system is correct? System I: $x + y \leq 12$ and $2x + 3y \leq 30$. System II: $x + y \leq 12$ and $3x + 2y \leq 30$.

- A) I only B) II only C) Both I and II D) Neither I nor II
-

Question 18 — Skill: Literal Equations | Difficulty: Medium

$h = \frac{(5r)^2}{3s}$. Which equation correctly expresses r in terms of h and s ?

- A) $r = \frac{\sqrt{3hs}}{5}$ B) $r = 5\sqrt{3hs}$ C) $r = \sqrt{\frac{3hs}{5}}$ D) $r = \sqrt{3hs} - 5$
-

Question 19 — Skill: Coordinate Geometry | Difficulty: Medium

Line m passes through $(0, 0)$ and is parallel to $y = 5x + 3$. If m also passes through $(4, d)$, what is d ?

Question 20 — Skill: Circle Equations | Difficulty: Medium

A circle has center $(11, 23)$ and radius $9k$. Which equation represents this circle?

- A) $(x - 11)^2 + (y - 23)^2 = 9k$ B) $(x - 11)^2 + (y - 23)^2 = 81k^2$
C) $(x - 11)^2 + (y - 23)^2 = 9k^2$ D) $(x - 11)^2 + (y - 23)^2 = 81k$
-

Question 21 — Skill: Right Triangle Trigonometry | Difficulty: Hard

[Figure: A right triangle with angle θ at the base, vertical leg = height of ramp, horizontal leg = x .] A ramp must have $\tan \theta \leq \frac{1}{10}$. If the height is 24.5 inches, what is the least possible value of x , in inches?

Question 22 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[Graph: Exponential curve $y = a^x + b$ passing through $(0, -3)$ with horizontal asymptote $y = -4$.] What is the value of $a - b$?

SECTION 2: MATH**MODULE 2 — EASIER PATH (22 Questions)**

For students scoring below ~60% on Module 1 — ~70% Easy/Medium, ~30% Hard

Question 1 — Skill: Function Notation & Evaluation | Difficulty: Easy

$f(x) = \frac{x + 12}{4}$. What is $f(-4)$?

- A) -4 B) -1 C) 2 D) 4
-

Question 2 — Skill: Statistics: Mean & Median | Difficulty: Easy

15, 18, 20, 22, 24, 25, 31. What is the mean?

- A) 20 B) 22 C) 24 D) 25
-

Question 3 — Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

[Graph: Decreasing exponential, starting at 2°C at $t = 0$.] At what time (minutes) is the temperature 10% of its initial value?

- A) 0.5 B) 1.0 C) 4.0 D) 2.3
-

Question 4 — Skill: Linear & Exponential Models from Data | Difficulty: Medium

[Graph: Scatterplot with positive-slope line, y -intercept ~ 8 .] Data set F adds 5 to each y -value. What is the y -intercept for data set F?

- A) 3 B) 8 C) 13 D) 18
-

Question 5 — Skill: Graphs of Linear Equations | Difficulty: Easy

$g(0) = 8$ and $g(4) = 8$. Which equation defines g ?

- A) $g(x) = 0$ B) $g(x) = 4$ C) $g(x) = 8$ D) $g(x) = x + 8$
-

Question 6 — Skill: Graphs of Linear Equations | Difficulty: Medium

[Table: $x \mid f(x): 2 \mid 0, ; 6 \mid -20, ; 12 \mid -50$.] Which equation defines f ?

- A) $f(x) = -5(x - 2)$ B) $f(x) = -5(5x - 3)$ C) $f(x) = -5(x + 2)$ D) $f(x) = -5(x - 4)$
-

Question 7 — Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

$m(t) = 200(0.5)^{t/15}$ models a 200g sample's decay (half-life 15 years). Best interpretation of $m(45) = 25$?

- A) Approximately 25 grams remains 45 years after decay starts.
B) The mass decreased by ~25 grams 45 years after decay starts.
C) The mass decreased by ~45 grams 25 years after decay starts.
D) Approximately 45 grams remains 25 years after decay starts.
-

Question 8 — Skill: Conditional Probability & Two-Way Tables | Difficulty: Medium

[Table: Engineering 40%, Marketing 25%, Sales 20%, HR 15%.] $P(\text{HR} \mid \text{NOT Engineering})$ is closest to?

- A) 0.15 B) 0.60 C) 0.38 D) 0.25
-

Question 9 — Skill: Volume & Surface Area | Difficulty: Medium

A right circular cone has base diameter 18 inches and height 10 inches. Volume = $k\pi$ cubic inches. What is k ?

Question 10 — Skill: Triangle Properties | Difficulty: Medium

In triangle PQR , S is the midpoint of PQ and T is the midpoint of QR . If $ST = 6$ and $PT = 11$, what is the length of PR ?

- A) 22 B) 17 C) 12 D) 6
-

Question 11 — Skill: Linear Equations in One & Two Variables | Difficulty: Medium

$x(r - 5) + 4 = 17x + 30$. If this equation has exactly one solution, what CANNOT be r ?

- A) 4 B) 10 C) 18 D) 22

Question 12 — Skill: Ratios, Rates & Proportions | Difficulty: Easy

In a jar: 15% red, 20% green, 40% blue, 25% white. If there are 30 green marbles, how many blue marbles are there?

- A) 20 B) 40 C) 60 D) 75
-

Question 13 — Skill: Quadratic Equations - Solving | Difficulty: Hard

$x(11x + 23) + c = 7$. A solution is $\frac{-23 + \sqrt{309}}{22}$. What is c ?

- A) 12 B) 7 C) 5 D) 3
-

Question 14 — Skill: Factoring Polynomials | Difficulty: Hard

One factor of $3x^3 + 33x^2 + 168x$ is $x + d$ (where d is positive). What is the smallest possible value of d ?

Question 15 — Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A model: deer population increases 150% each year from 2018 to 2023. At the end of 2019, there were 375 deer. Which equation gives n (deer) as a function of t (years after 2018)?

- A) $n = 150(1.5)^t$ B) $n = 150(2.5)^t$ C) $n = 375(1.5)^t$ D) $n = 375(2.5)^t$
-

Question 16 — Skill: Area & Perimeter | Difficulty: Medium

[Figure: A $20 \text{ ft} \times 15 \text{ ft}$ garden with a w -ft-wide gravel border on all sides, where the border area = 200 ft^2 .] What is w ?

- A) 2.5 B) 5 C) 18 D) 36
-

Question 17 — Skill: Creating Linear Models from Word Problems | Difficulty: Hard

[Table: x | y : $-3t$ | 16, ; $-t$ | 12, ; t | 8.] Which equation represents the linear relationship?

- A) $tx + 2y = 10t$ B) $2x + ty = 10t$ C) $2x + ty = 10$ D) $tx + 2y = 10$

Question 18 — Skill: Linear Inequalities | Difficulty: Hard

In five consecutive even integers (ordered least to greatest), the first is x . The product of 36 and the third integer is at most 60 less than the sum of the first and fifth integers. What is the greatest possible value of x ?

Question 19 — Skill: Ratios, Rates & Proportions | Difficulty: Hard

A blueprint has a 24-inch side where 1 inch = 12 feet. A reduced copy has sides 50% shorter. On the reduced copy, 1 inch represents how many feet?

- A) 6.00 B) 8.50 C) 17.00 D) 24.00

Question 20 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

$p(x) = a(3.2)^{x+b}$, $q(x) = a(3.2)^x + b$. Given $a < b$ and $b < 0$. Which equation displays the y -intercept as a constant or coefficient?

- A) p only B) q only C) Both p and q D) Neither p nor q

Question 21 — Skill: Function Notation & Evaluation | Difficulty: Hard

$k(x) = -\sqrt{x^2 + bx + c}$. The graph contains $(3, 0)$ and $(0, -\sqrt{195})$. If $k(m) = 0$, what is the greatest possible value of m ?

Question 22 — Skill: Function Transformations | Difficulty: Hard

$g(x) = \frac{|x|}{a} - 18$, where $a < 0$. What is the product of $g(12a)$ and $g(6a)$?

SECTION 2: MATH**MODULE 2 — HARDER PATH (22 Questions)**

For students scoring above ~60% on Module 1 — ~30% Easy/Medium, ~70% Hard

Question 1 — Skill: Function Notation & Evaluation | Difficulty: Medium

$g(x) = \frac{2x + 16}{5}$. What is $g(-3)$?

- A) -2 B) 0 C) 2 D) 4

Question 2 — Skill: Statistics: Mean & Median | Difficulty: Easy

12, 14, 17, 19, 21, 22, 28. What is the mean?

- A) 17 B) 19 C) 21 D) 22

Question 3 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[Graph: Radioactive decay, starting at 800 becquerels at $t = 0$.] At what time (hours) is activity 25% of initial?

- A) 1.0 B) 1.5 C) 4.5 D) 2.8

Question 4 — Skill: Linear & Exponential Models from Data | Difficulty: Hard

GRAPH TO INSERT

[Graph: Negative-slope line, y -intercept ~ 72 .] Data set S subtracts 11 from each y -value. What is the y -intercept for data set S?

- A) 50 B) 55 C) 61 D) 66

Question 5 — Skill: Graphs of Linear Equations | Difficulty: Medium

$h(0) = 12$ and $h(6) = 12$. Which equation defines h ?

- A) $h(x) = 0$ B) $h(x) = 6$ C) $h(x) = 12$ D) $h(x) = x + 12$

Question 6 — Skill: Graphs of Linear Equations | Difficulty: Hard

[Table: x | $f(x)$: 3 | -30, ; 7 | -54, ; 14 | -96.] Which equation defines f ?

- A) $f(x) = -6(x - 5) - 6(x + 1) + 4$ B) $f(x) = -6(6x - 2)$ C) $f(x) = -6(x + 2)$ D) $f(x) =$
-

Question 7 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

$d(t) = 500(0.5)^{t/40}$ models 500g decay (half-life 40 years). Best interpretation of $d(120) = 62.5$?

- A) Approximately 62.5 grams remains 120 years after decay starts.
B) The mass decreased by ~62.5 grams 120 years after decay starts.
C) The mass decreased by ~120 grams 62.5 years after decay starts.
D) Approximately 120 grams remains 62.5 years after decay starts.
-

Question 8 — Skill: Conditional Probability & Two-Way Tables | Difficulty: Hard

[Table: Freshman 30%, Sophomore 25%, Junior 24%, Senior 21%.] $P(\text{Senior} \mid \text{at least Junior})$ is closest to?

- A) 0.21 B) 0.35 C) 0.47 D) 0.55
-

Question 9 — Skill: Volume & Surface Area | Difficulty: Hard

Right circular cylinder: base diameter 20 inches, height 9 inches. Volume $= k\pi$. What is k ?

Question 10 — Skill: Triangle Properties | Difficulty: Hard

In triangle XYZ , M is the midpoint of XY , N is the midpoint of YZ . $MN = 11$, $MZ = 18$. What is the length of XZ ?

- A) 36 B) 29 C) 22 D) 18
-

Question 11 — Skill: Linear Equations in One & Two Variables | Difficulty: Hard

$x(r - 8) + 5 = 23x + 40$. If this equation has exactly one solution, what CANNOT be r ?

- A) 5 B) 12 C) 20 D) 31

Question 12 — Skill: Quadratic Equations - Solving | Difficulty: Hard

$x(15x + 31) + c = 8$. A solution is $\frac{-31 + \sqrt{481}}{30}$. What is c ?

- A) 16 B) 12 C) 8 D) 4
-

Question 13 — Skill: Factoring Polynomials | Difficulty: Hard

One factor of $4x^3 + 52x^2 + 264x$ is $x + d$ (where d is positive). What is the smallest possible value of d ?

Question 14 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

An investment increases 120% each year, 2020–2025. At the end of 2021: \$440. Which equation gives v (value) as a function of t (years after 2020)?

- A) $v = 200(1.2)^t$ B) $v = 200(2.2)^t$ C) $v = 440(1.2)^t$ D) $v = 440(2.2)^t$
-

Question 15 — Skill: Area & Perimeter | Difficulty: Hard

[Figure: A 16 in. $\times$ 12 in. photo with a w -inch frame on all sides. Frame area = 288 in².] What is w ?

- A) 3 B) 12 C) 8 D) 4
-

Question 16 — Skill: Creating Linear Models from Word Problems | Difficulty: Hard

[Table: x | y : $-4t$ | 18, ; $-2t$ | 12, ; $2t$ | 0.] Which equation represents the linear relationship?

- A) $tx + 3y = 6t$ B) $3x + ty = 6t$ C) $3x + ty = 6$ D) $tx + 3y = 6$
-

Question 17 — Skill: Linear Inequalities | Difficulty: Hard

In six consecutive odd integers (least to greatest), the first is x . The product of 30 and the fourth integer is at most 42 less than the sum of the first and sixth integers. What is the greatest possible value of x ?

Question 18 — Skill: Ratios, Rates & Proportions | Difficulty: Hard

A floor plan has 40-inch sides where 1 inch = 20 feet. A reduced copy has sides 70% shorter. On the reduced version, 1 inch represents approximately how many feet?

- A) 6.00 B) 14.00 C) 42.86 D) 66.67
-

Question 19 — Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

$f(x) = a(4.5)^{x+b}$, ; $g(x) = a(4.5)^x + b$. Given $a < b$ and $b < 0$. Which equation displays the y -intercept as a constant or coefficient?

- A) f only B) g only C) Both f and g D) Neither f nor g
-

Question 20 — Skill: Function Notation & Evaluation | Difficulty: Hard

$j(x) = -\sqrt{x^2 + bx + c}$. The graph contains $(5, 0)$ and $(0, -\sqrt{510})$. If $j(m) = 0$, what is the greatest possible value of m ?

Question 21 — Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

$f(x) = \frac{|x|}{a} - 22$, where $a < 0$. What is $f(10a) \times f(5a)$?

Question 22 — Skill: Quadratic Word Problems | Difficulty: Hard

If $3x^2 - 7x - 20 = 0$, and $x > 0$, what is $9x$?